Farhad Mohsin

fmohsin@holycross.edu | farhadmohsiniii@gmail.com +1 (508) 793-2872
College of the Holy Cross, 1 College St PO Box 48A, Worcester, MA-01610
https://scholar.google.com/citations?user=Akfw1iQAAAAJ
https://farhadmohsin.github.io

Experience

Sep 2023 - Present	**Dept. of Math and Computer Science, College of the Holy Cross** *Assistant Professor of Computer Science*
Aug 2018 - Jul 2023	**Dept. of Computer Science, Rensselaer Polytechnic Institute** *Graduate Research Assistant.* Advised by Prof. Lirong Xia. *Graduate Teaching Assistant*

Teaching

College of the Holy Cross

- `CSCI 307` Data Mining (Fall 2025, Fall 2023)
- `CSCI 335` Advanced Algorithms (Originated in Spring 2025)
- `CSCI 235` Analysis of Algorithms (Fall 2024, Spring 2024)
- `CSCI 132` Data Structures (Spring 2025, Spring 2024, Fall 2023)
- `CSCI 135` Discrete Structures (Fall 2024)
- `CSCI 131` Techniques of Programming (Fall 2025)
- Competitive Programming Contest Coaching (Fall 2024)
- Honors thesis supervision:
  - Matthew Kalarickal, Exploring Communication in Multi-agent Cooperative Reinforcement Learning (2025, CS Honors Thesis)
  - William Schimitsch, Achieving Fairness in Zoning Laws with Machine Learning (2025, College Honors Thesis)

Rensselaer Polytechnic Institute

- `CSCI 4150` Introduction to Artificial Intelligence (Lead TA in Spring 2023; TA in Spring 2021, 2022)
- `CSCI 1200` Data Structures (TA in Fall 2018)

Research

Research Interests

- **Computational Social Choice:** Investigating the axiomatic properties and structural vulnerabilities (e.g., voting paradoxes) of collective decision-making, while developing interactive visualization tools.
- **Preference Learning & AI-aided Decision-making:** Leveraging theory with modern machine learning techniques and Large Language Models (LLMs) to unobtrusively elicit, estimate, and aggregate human preferences.

- **Data Mining & Algorithmic Fairness:** Utilizing machine learning techniques and explainable AI (XAI) frameworks to evaluate, optimize, and interpret fairness in complex social systems.

Institutional Research Funding & Activity

- **Fall 2025, Spring 2026:** Recipient, Holy Cross Research Associate Grant for AI-aided collective decision-making project.
- **Summer 2025:** Faculty Advisor, Weiss Summer Research Program
  Students worked on projects on: *Illustrated Social Choice* and *Estimating PL Parameters Using Neural Networks*.

Research Publications

Ongoing Work

1. **Mohsin**, **F.** (2026). The No-show Paradox in Single Transferable Vote under One-dimensional Preferences. *arXiv preprint arXiv:2606.12785.*

Journal Papers

2. **Mohsin**, **F.**, Liu, A., Chen, P.-Y., Rossi, F., & Xia, L. (2022). Learning to design fair and private voting rules. *Journal of Artificial Intelligence Research*, *75*, 1139–1176.

Conference Proceedings

3. **Mohsin**, **F.**, Han, Q., Ruan, S., Chen, P.-Y., Rossi, F., & Xia, L. (2024). Computational Complexity of Verifying the Group No-Show Paradox [previously appeared as an extended abstract in AAMAS-2023]. *33rd International Joint Conference on Artificial Intelligence (IJCAI-24).*

4. Liu, S., **Mohsin**, **F.**, Xia, L., & Seneviratne, O. (2019). Strengthening Smart Contracts to Handle Unexpected Situations. *2019 IEEE International Conference on Decentralized Applications and Infrastructures (DAPPCON)*, 182–187.

5. Hasan, S. M., Monjil, M. B., **Mohsin**, **F.**, Hayat, M. A., & Rashid, A. B. M. H.-u. (2015). Adaptive Beamforming with a Microphone Array. *2015 18th International Conference on Computer and Information Technology (ICCIT)*, 178–183.

Peer-reviewed Workshop Papers

6. Schimitsch, W., & **Mohsin**, **F.** (2025). Achieving Fairness in Zoning Laws with Machine Learning. *Third IJCAI Workshop on Computational Fair Division.*

7. **Mohsin**, **F.**, Kang, I., Chen, P.-Y., Rossi, F., & Xia, L. (2025). Learning Individual and Collective Priorities over Moral Dilemmas. *Social Choice for AI Ethics and Safety 2025 (SC4AI'25) at AAMAS 2025.*

8. **Mohsin**, **F.**, Kang, I., Chen, Y., Shang, J., & Xia, L. (2023). Dependency and Coreference-boosted Multi-Sentence Preference model. *The Ninth International Workshop on Deep Learning on Graphs: Method and Applications (DLG-AAAI-23).*

9. **Mohsin**, **F.**, Luo, L., Ma, W., Kang, I., Zhao, Z., Liu, A., Vaish, R., & Xia, L. (2021). Making Group Decisions from Natural Language-Based Preferences. *The 8th International Workshop on Computational Social Choice (COMSOC-2021).*

10. **Mohsin**, **F.**, Zhao, X., Hong, Z., de Mel, G., Xia, L., & Seneviratne, O. (2019). Ontology Aided Smart Contract Execution for Unexpected Situations. *Proceedings of the Blockchain enabled Semantic Web Workshop (BlockSW) and Contextualized Knowledge Graphs (CKG) Workshop.*

Technical Reports

11. Kang, I., Ruan, S., Ho, T., Lin, J.-C., **Mohsin**, **F.**, Seneviratne, O., & Xia, L. (2023). LLM-augmented Preference Learning from Natural Language. *arXiv preprint arXiv:2310.08523.*

12. Han, Q., Ruan, S., Kong, Y., Liu, A., **Mohsin**, **F.**, & Xia, L. (2021). Truthful Information Elicitation from Hybrid Crowds. *arXiv preprint arXiv:2107.10119.*

Books

13. Hasan, M., Roy, A., Chakraborty, T., & **Mohsin**, **F.** (2014). *BdMO Prostuti* [This is a Math Olympiad problem-solving textbook in Bengali targeted at High School Students]. Tamrolipi Publishers.

Education

2018 – 2023 **Ph.D. in Computer Science**
Rensselaer Polytechnic Institute (RPI), Troy, NY

2010 – 2015 **B.Sc. in Electrical and Electronic Engineering**
Bangladesh University of Engineering and Technology (BUET), Bangladesh

Industry Experience

Sep 2015 - Aug 2018 **Grameenphone Ltd (Telenor Group), Dhaka, Bangladesh**
Telecommunications Engineer at Service Assurance and Security

- Developed tools for automating resource planning and auditing in telecommunication network and efficient database management using SQL and Python.
- Engineered Machine Learning algorithms for efficient and automated fault detection in high dimensional Radio Access Network (RAN) data.

Skills

Languages	Python, Java, C/C++, R, MATLAB
Machine Learning	PyTorch, PyTorch Geometric, scikit-learn, XGBoost, Stable-Baselines3
LLM Frameworks	DSPy, Ollama, Hugging Face Transformers
Optimization	Gurobi, CVXPY, CVXOPT
Tools & Concepts	p5.js, MPI (Parallel Computing), SQL, Data-Intensive Systems Foundations